

FALL 2026 | MEE4033

Mechatronics

Mechatronics (MEE4033) · Department of Mechanical Engineering

Lecture 5

AC Network Analysis 2

Principles and Applications of Electrical Engineering, 7th ed., Chapter 3
Electric Circuits, 11th ed., Chapter 9

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3.4 Phasor Solution of Circuits with Sinusoidal Sources

Sinusoidal signals can be represented as complex numbers.

- Time domain signal representation.

$$v(t) = A \cos(\omega t + \theta)$$

- Frequency domain signal representation.

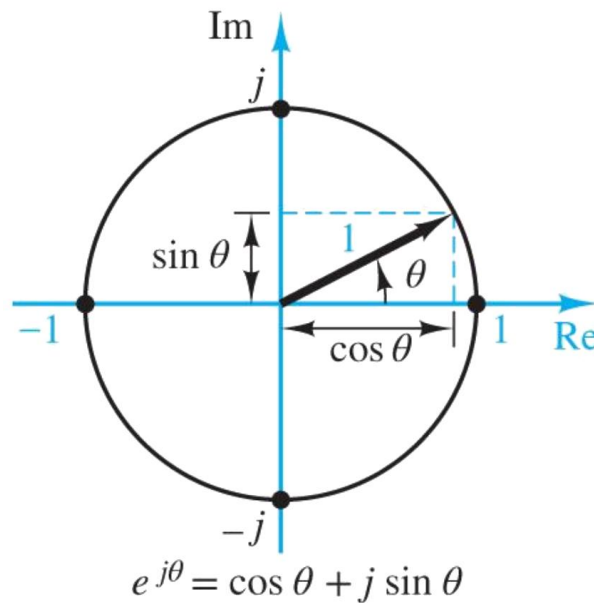
$$V(j\omega) = Ae^{j\theta} = A\angle\theta = A(\cos\theta + j\sin\theta)$$

Euler's Formula

- Euler's Formula

$$e^{j\theta} = \cos\theta + j\sin\theta$$

- Complex exponential represented as a unit phasor with real and imaginary components.



$$e^{j\theta} = \underline{\angle\theta}$$

Unit phasor angle θ

Polar and Rectangular Forms of Common Phasors

Complex exponential	Polar	Rectangular
$Ae^{\pm j(\pi/6)}$	$A\angle \pm \pi/6$	$A(\sqrt{3}/2 \pm j/2)$
$Ae^{\pm j\pi/4}$	$A\angle \pm \pi/4$	$A(\sqrt{2}/2 \pm j\sqrt{2}/2)$
$Ae^{\pm j\pi/3}$	$A\angle \pm \pi/3$	$A(1/2 \pm j\sqrt{3}/2)$
$Ae^{\pm j \arctan(3/4)}$	$A\angle \pm \arctan(3/4)$	$A(0.8 \pm j0.6)$
$Ae^{\pm j \arctan(4/3)}$	$A\angle \pm \arctan(4/3)$	$A(0.6 \pm j0.8)$

Phasors

- Sinusoidal waveform is expressed as complex numbers

$$\cos(\omega t + \theta) = \operatorname{Re}(Ae^{j(\omega t + \theta)})$$

- The expression can be simplified without explicit expression of the terms $e^{j\omega t}$ and Re

Why? Frequency (ω) is common to all voltages and currents

$$\operatorname{Re}\{Ae^{j(\omega t + \theta)}\} = \operatorname{Re}\{Ae^{j\omega t} e^{j\theta}\} \Rightarrow Ae^{j\theta}$$

- The symbol $\Rightarrow$ indicates equality with $e^{j\omega t}$ and Re hidden but understood implicitly.

Phasors

- **Phasors: A complex number in polar form, with a magnitude and phase angle**
- $A e^{j\theta}$ is a complex number that carries the amplitude and phase angle of the cosine function

PHASOR TRANSFORM

$$\mathbf{V} = V_m e^{j\phi} = \mathcal{P} \{ V_m \cos(\omega t + \phi) \}$$

- Phasor transform transfers the sinusoidal function from the time domain to complex-number domain

Phasors

- The complex-number domain is also called the frequency domain, since the response depends, in general, on ω (not on time)

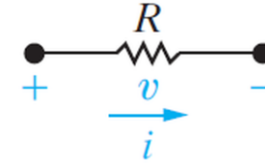
$$\mathbf{V} = V_m e^{j\phi} = \mathcal{P} \{ V_m \cos(\omega t + \phi) \}$$
 Polar form of a phasor

$$\mathbf{V} = V_m \cos \phi + jV_m \sin \phi$$
 Rectangular form of a phasor

$$A \angle \phi^\circ \equiv A e^{j\phi}$$
 Angle notation

Impedance of a Resistor

- V-I relationship for a Resistor



- If the current in a resistor is: $i = I_m \cos(\omega t + \theta_i)$
- The voltage at the terminals of the resistor is:

$$\begin{aligned} v &= R[I_m \cos(\omega t + \theta_i)] \\ &= RI_m[\cos(\omega t + \theta_i)] \end{aligned}$$

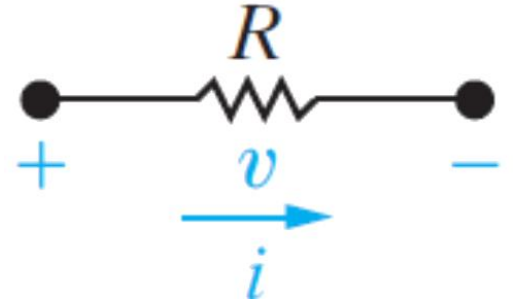
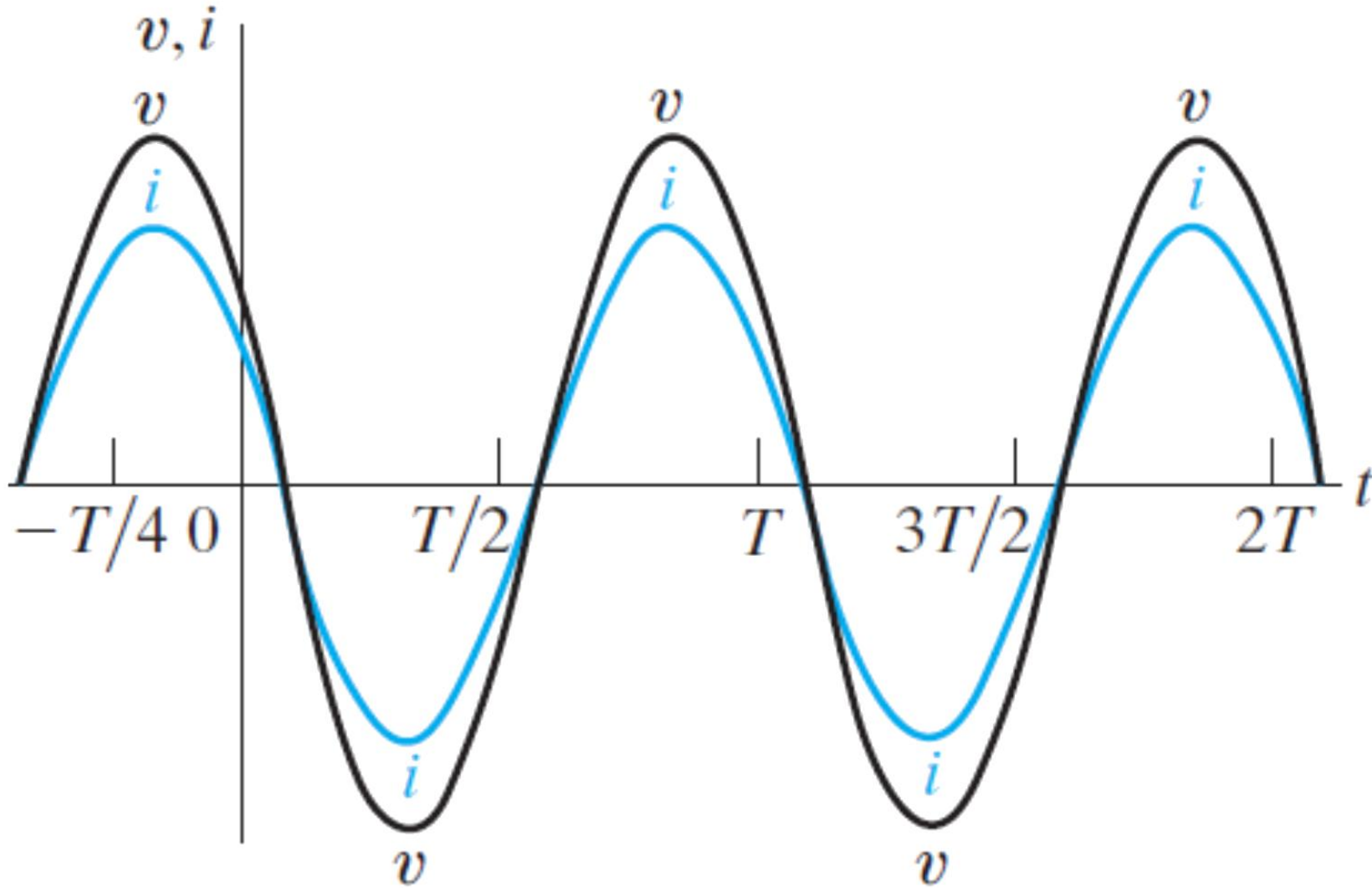
- The phasor transform of this voltage is:

$$\mathbf{V} = RI_m e^{j\theta_i} = RI_m \angle \theta_i \mathbf{I}$$

RELATIONSHIP BETWEEN PHASOR VOLTAGE AND PHASOR CURRENT FOR A RESISTOR

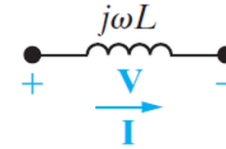
$$\mathbf{V} = R\mathbf{I}, \quad (9.10)$$

Impedance of an Inductor



Impedance of an Inductor

- V-I relationship for an inductor



- If the current in an inductor is: $i = I_m \cos(\omega t + \theta_i)$
- The voltage at the terminals of the inductor is:

$$\begin{aligned} v &= L \frac{di}{dt} = -\omega L I_m \sin(\omega t + \theta_i) \\ &= -\omega L I_m \cos(\omega t + \theta_i - 90^\circ) \end{aligned}$$

- The phasor transform of this voltage is:

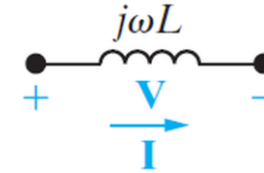
$$\begin{aligned} \mathbf{V} &= -\omega L I_m e^{j(\theta_i - 90^\circ)} \\ &= -\omega L I_m e^{j\theta_i} e^{-j90^\circ} \\ &= j\omega L I_m e^{j\theta_i} \\ &= j\omega L I_m \angle \theta_i. \end{aligned}$$

RELATIONSHIP BETWEEN PHASOR VOLTAGE AND PHASOR CURRENT FOR AN INDUCTOR

$$\mathbf{V} = j\omega L \mathbf{I}. \quad (9.11)$$

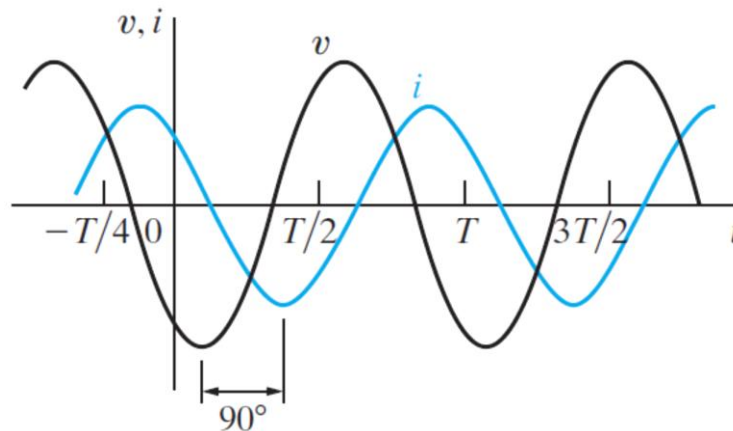
Impedance of an Inductor

- We can rewrite the equation:



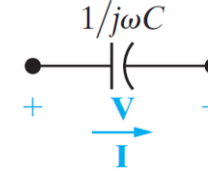
$$\begin{aligned}\mathbf{V} &= (\omega L \angle 90^\circ) I_m \angle \theta_i \\ &= \omega L I_m \angle (\theta_i + 90)^\circ,\end{aligned}$$

- which indicates that the voltage and current are out of phase by exactly 90 degrees



Impedance of a Capacitor

- V-I relationship for a capacitor



- If the voltage across a capacitor is:

$$v = V_m \cos(\omega t + \theta_v)$$

- The current through the capacitor is:

$$\begin{aligned} i &= C \frac{dv}{dt} = -\omega C V_m \sin(\omega t + \theta_v) \\ &= -\omega C V_m \cos(\omega t + \theta_v - 90^\circ) \end{aligned}$$

- The phasor transform of this current is:

$$\begin{aligned} \mathbf{I} &= -\omega C V_m e^{j(\theta_v - 90^\circ)} \\ &= -\omega C V_m e^{j\theta_v} e^{-j90^\circ} \\ &= j\omega C V_m e^{j\theta_v} \mathbf{V} \\ &= j\omega C V_m \angle \theta_v \end{aligned}$$

RELATIONSHIP BETWEEN PHASOR VOLTAGE AND PHASOR CURRENT FOR A CAPACITOR

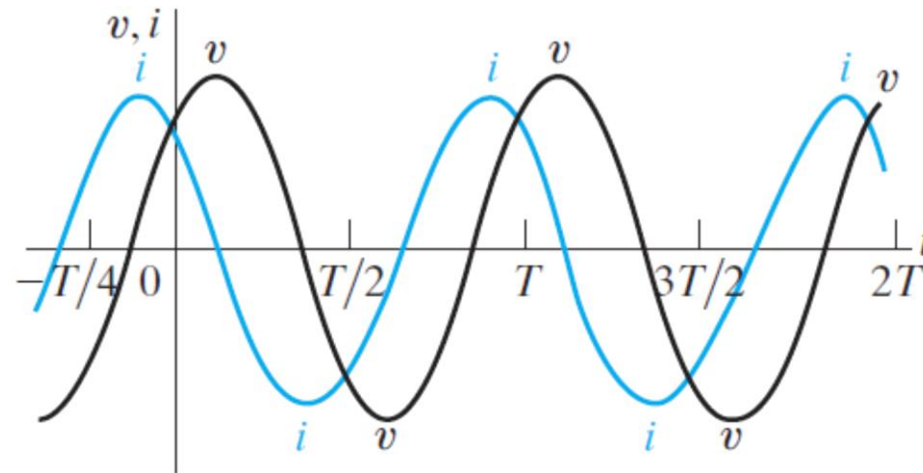
$$\mathbf{V} = \frac{1}{j\omega C} \mathbf{I}. \quad (9.12)$$

Impedance of a Capacitor

- We can rewrite the equation:

$$\begin{aligned}\mathbf{V} &= \left(\frac{1}{\omega C} \angle -90^\circ \right) I_m \angle \theta_i \\ &= \frac{I_m}{\omega C} \angle (\theta_i - 90)^\circ.\end{aligned}$$

- which indicates that the current leads the voltage by 90 degrees



3.5 Impedance

i-v relationships of resistors, inductors, and capacitors can be expressed in phasor notation.

- Phasor quantity Z known as impedance.

$$\mathbf{Z}_R = R \qquad \mathbf{Z}_L = j\omega L \qquad \mathbf{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

- Admittance is the inverse of impedance.

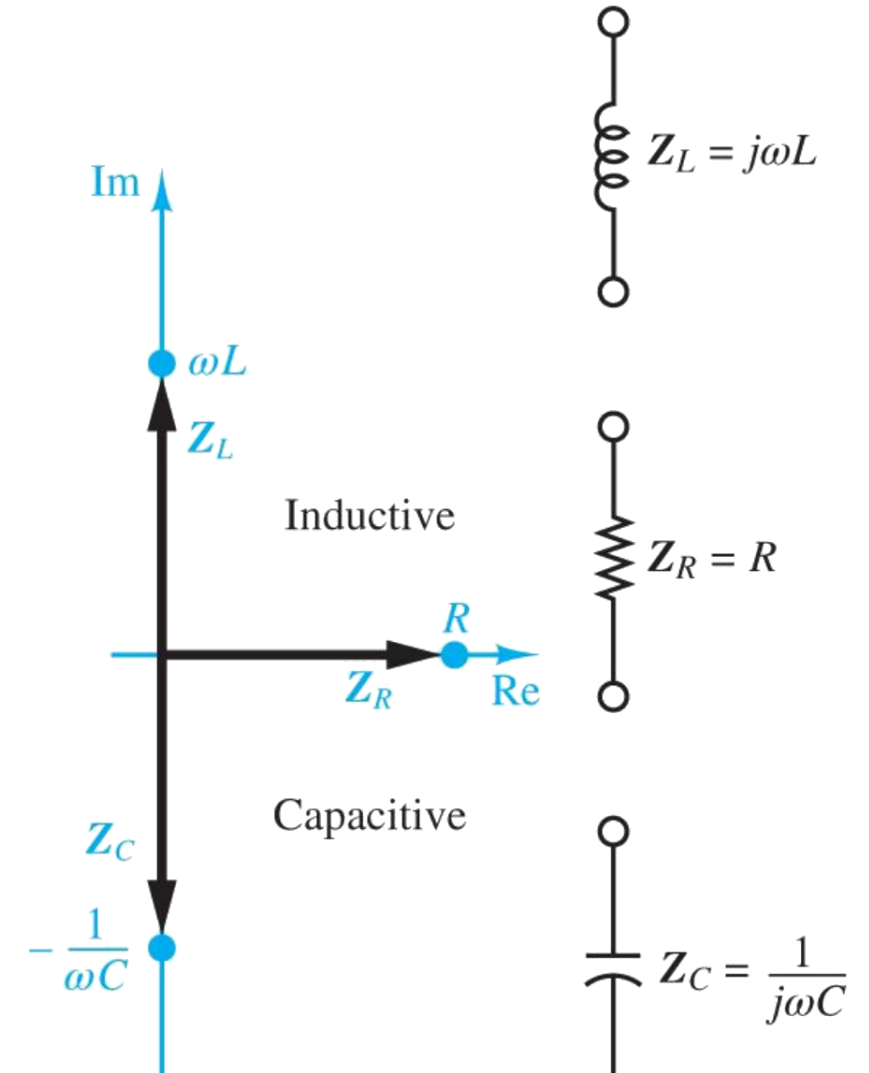
Generalized Ohm's Law: $V = I Z$

$$Z(j\omega) = R(j\omega) + jX(j\omega)$$

- R : resistance, X : reactance

Impedances in the Complex Plane

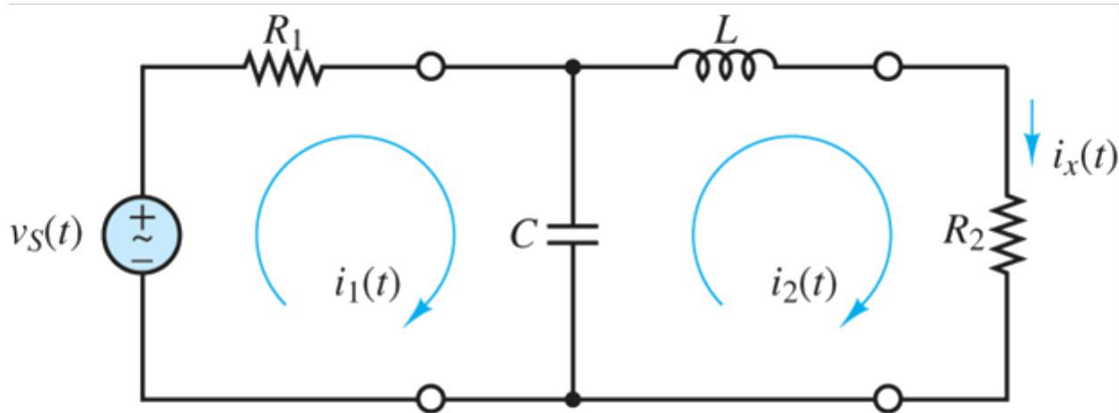
- The impedance concept is very useful in solving AC circuit analysis problems.
- It allows network theorems developed for DC circuits to be applied to AC circuits.
- $Z_R(j\omega)$, $Z_L(j\omega)$, and $Z_C(j\omega)$ in the complex plane



3.6 AC Circuit Analysis

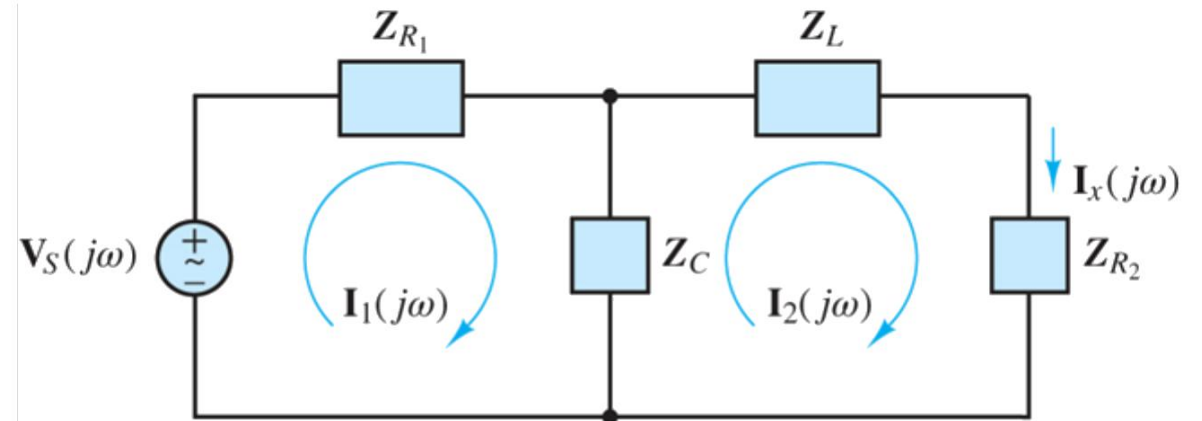
Phasors and impedance simplify AC circuit analysis.

- Allow use of same solution methods as DC circuits.



A sample circuit
for AC analysis

Time-domain form



The same circuit
in phasor form

Phasor-impedance form

3.6 AC Circuit Analysis

KVL in the frequency domain

$$v_1 + v_2 + \cdots + v_n = 0$$

$$V_{m_1} \cos(\omega t + \theta_1) + V_{m_2} \cos(\omega t + \theta_2) + \cdots + V_{m_n} \cos(\omega t + \theta_n) = 0$$

$$\mathcal{R}\{V_{m_1}e^{j\theta_1}e^{j\omega t}\} + \mathcal{R}\{V_{m_2}e^{j\theta_2}e^{j\omega t}\} + \cdots + \mathcal{R}\{V_{m_n}e^{j\theta_n}e^{j\omega t}\} = 0$$

$$\mathcal{R}\{V_{m_1}e^{j\theta_1}e^{j\omega t} + V_{m_2}e^{j\theta_2}e^{j\omega t} + \cdots + V_{m_n}e^{j\theta_n}e^{j\omega t}\} = 0$$

$$\mathcal{R}\{(V_{m_1}e^{j\theta_1} + V_{m_2}e^{j\theta_2} + \cdots + V_{m_n}e^{j\theta_n})e^{j\omega t}\} = 0$$

$$\mathcal{R}\{(\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n)e^{j\omega t}\} = 0$$

But $e^{j\omega t} \neq 0$

KVL IN THE FREQUENCY DOMAIN

$$\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n = 0,$$

3.6 AC Circuit Analysis

KCL in the frequency domain

KCL IN THE FREQUENCY DOMAIN

$$\mathbf{I}_1 + \mathbf{I}_2 + \cdots + \mathbf{I}_n = 0,$$

3.6 AC Circuit Analysis

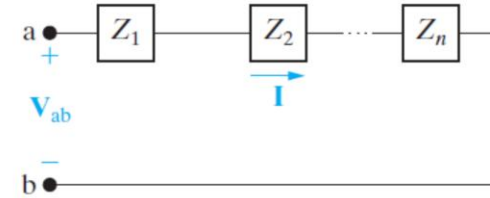
Impedances in Series

COMBINING IMPEDANCES IN SERIES

$$Z_{ab} = \frac{V_{ab}}{I} = Z_1 + Z_2 + \cdots + Z_n.$$

VOLTAGE DIVISION IN THE FREQUENCY DOMAIN

$$V_j = \frac{Z_j}{Z_{eq}} V_s. \quad (9.17)$$



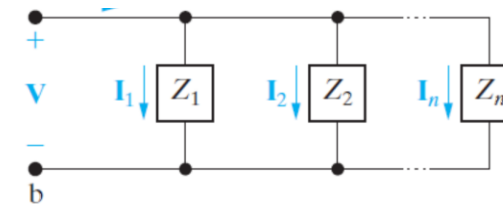
Impedances in Parallel

COMBINING IMPEDANCES IN PARALLEL

$$\frac{1}{Z_{ab}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_n}.$$

CURRENT DIVISION IN THE FREQUENCY DOMAIN

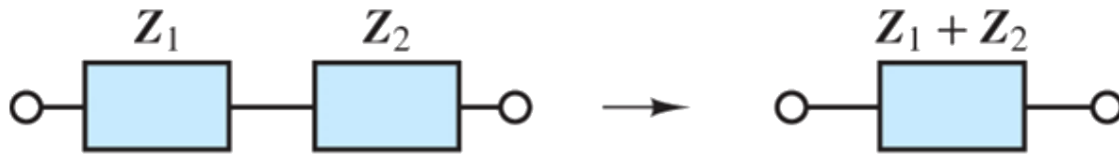
$$I_j = \frac{Z_{eq}}{Z_j} I_s. \quad (9.20)$$



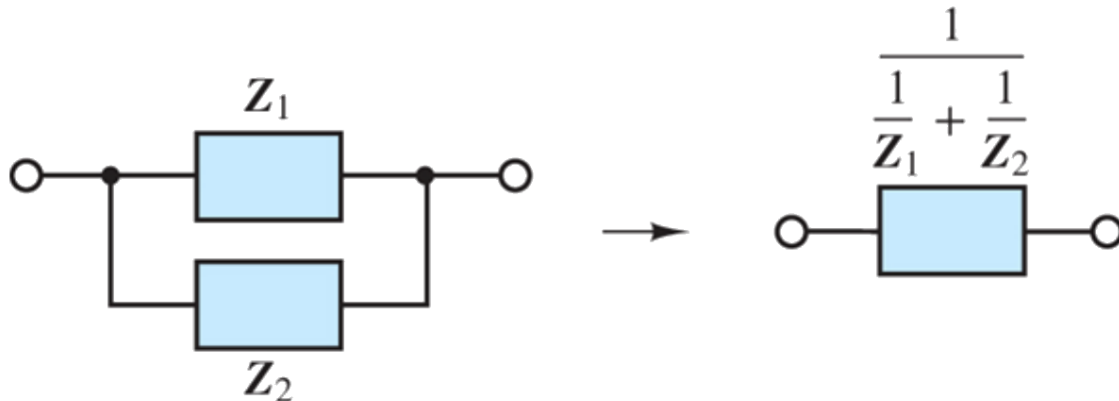
AC Equivalent Circuits

- Rules for impedance and admittance reduction.

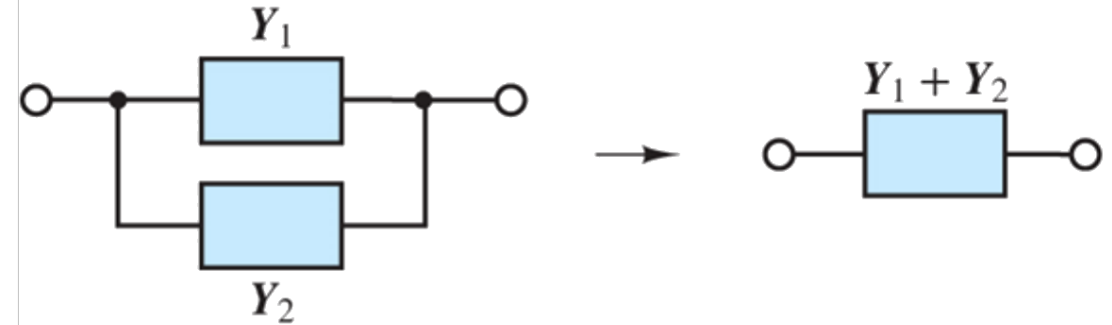
Impedances in series add:



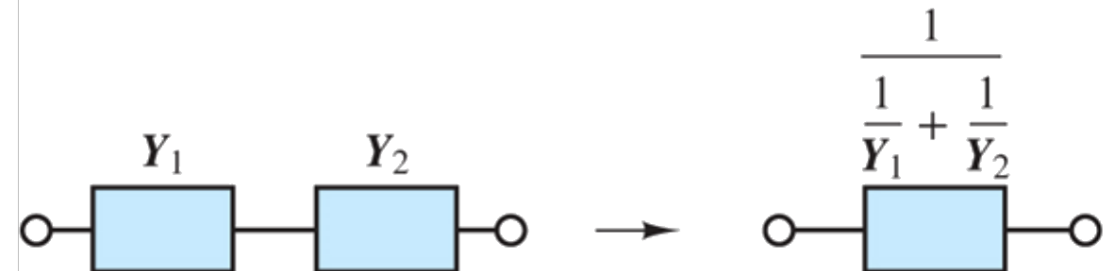
Impedances in parallel behave like resistors in parallel:



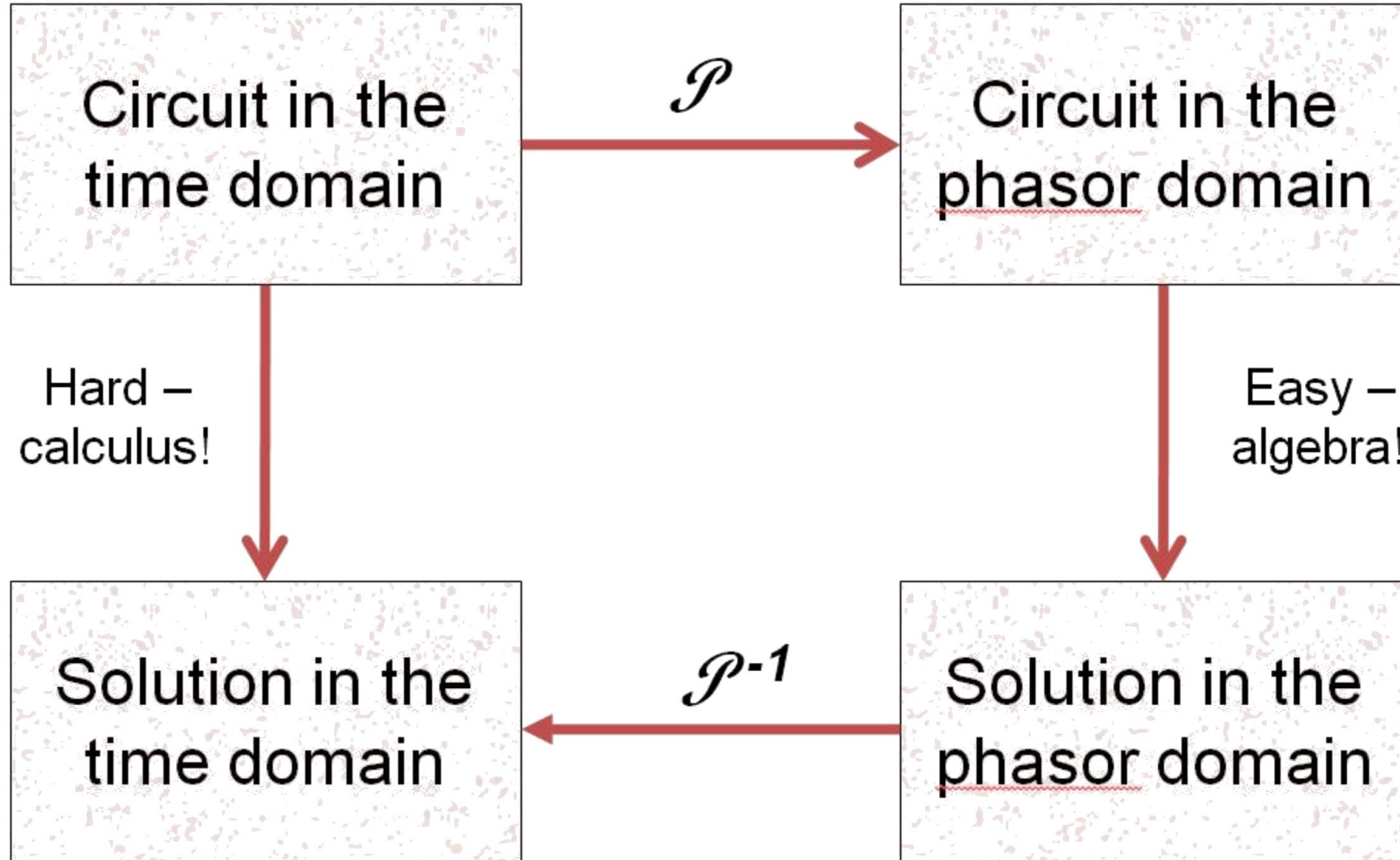
Admittances in parallel add:



Admittances in series behave like conductances in series:



Example 9.9: The Phasor Method at a Glance



Steps in AC Steady-State (ACSS) Analysis

Steps in ACSS (AC Steady State) Analysis:

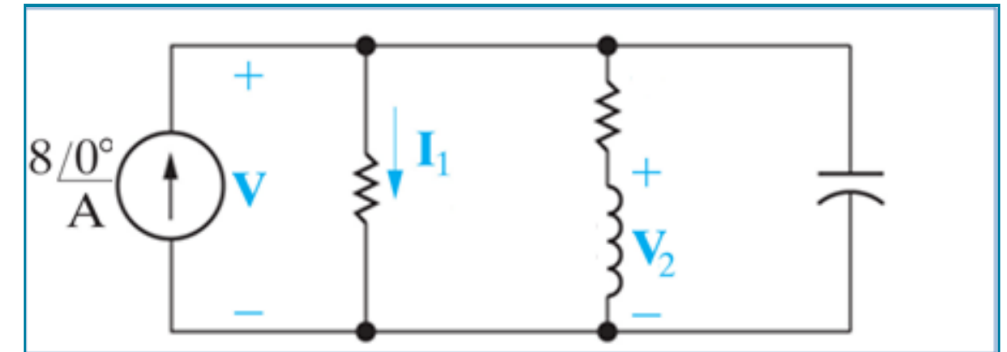
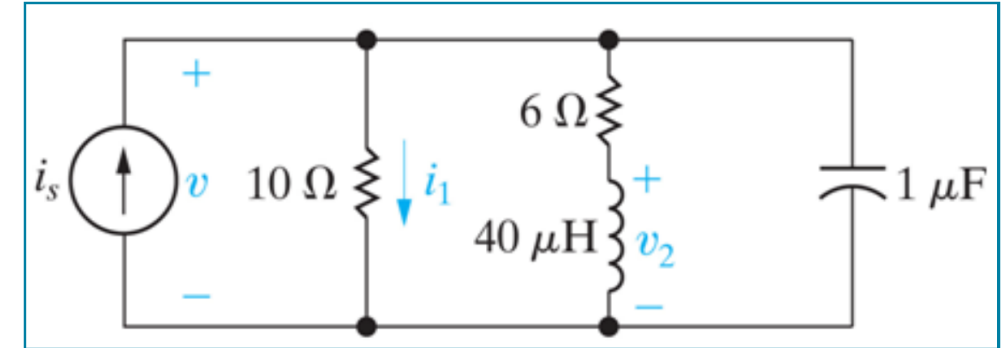
1. Redraw the circuit (the phasor transform does not change the components or their connections).
2. Phasor transform all known $v(t)$ and $i(t)$.
3. Represent unknown voltages and currents with V and I .
4. Replace component values with impedance (Z) values.
5. Use any circuit analysis method(s) to write equations and solve them with a calculator.
6. Inverse-transform the result, which is a phasor, back to the time domain.

Example 9.9

Example 9.9

$i_s(t) = 8 \cos(200,000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady state.

1. Redraw the circuit.
2. Phasor transform all known voltages and currents.
3. Represent unknown voltages and currents with phasor symbols



Example 9.9: Check Your Understanding (1)

The next step is to calculate the impedance of all resistors, inductors, and capacitors. Actually, the impedance of the resistors doesn't require calculation.

✓ A. True

✗ B. False

Example 9.9: Check Your Understanding (2)

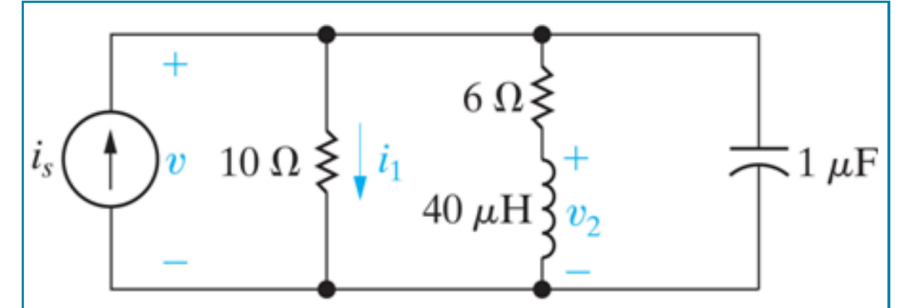
The impedance of inductors and capacitors is calculated using

-  A. The source frequency
-  B. The component value
-  C. Both A and B

Example 9.9: Step 4, Impedances

Example 9.9

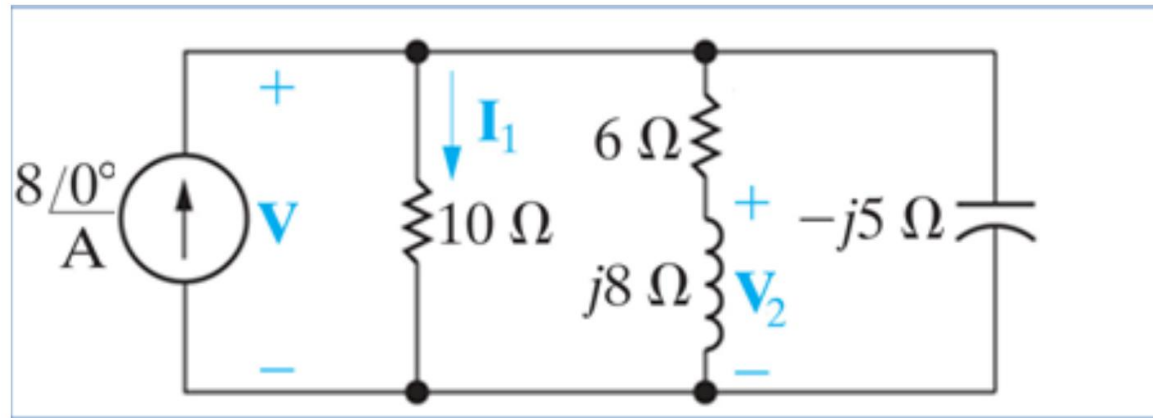
$i_s(t) = 8 \cos(200,000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady state.



4. Calculate the impedances of all resistors, inductors, and capacitors.

$$Z_L = j\omega L = j(200,000)(40\mu) = j8\Omega$$

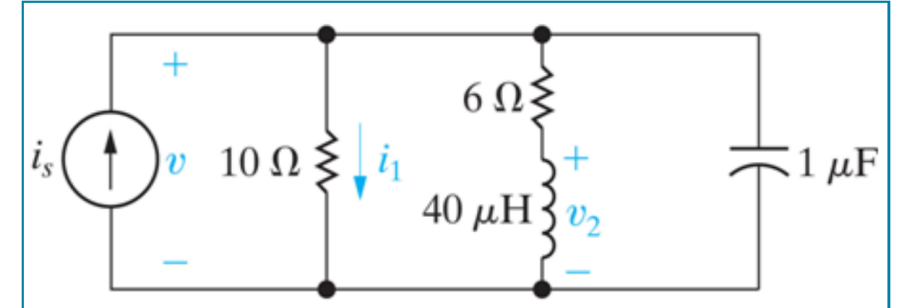
$$Z_C = \frac{-j}{\omega C} = \frac{-j}{(200,000)(1\mu)} = -j5\Omega$$



Example 9.9: Step 5, Find V

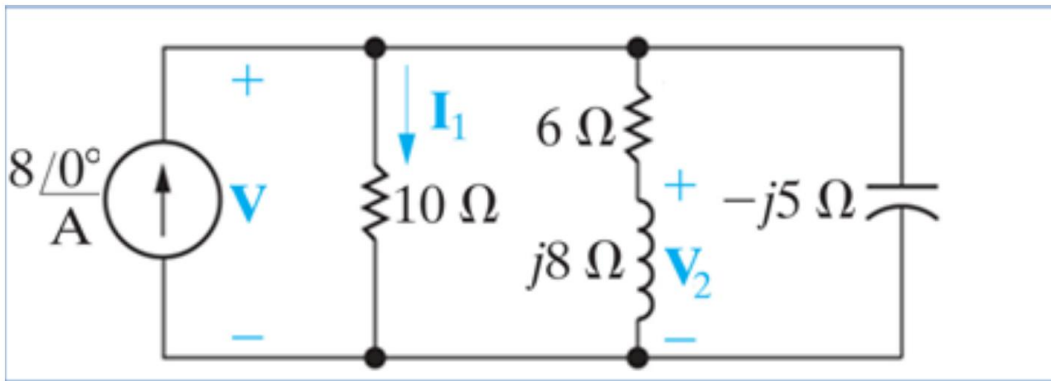
Example 9.9

$i_s(t) = 8 \cos(200,000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady state.



5. Use DC circuit analysis techniques to find the phasor V .

- In this problem, we'll simplify the circuit by replacing all impedances by one equivalent impedance, and use Ohm's law for phasors.



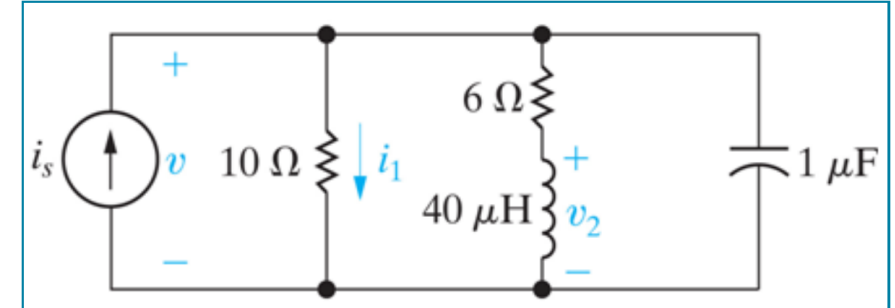
$$Z_{eq} = 10 \parallel (6 + j8) \parallel -j5 = (4 - j3)\Omega$$

$$\mathbf{V} = \mathbf{Z}\mathbf{I} = (4 - j3)(8\angle 0^\circ) = (32 - j25)V$$

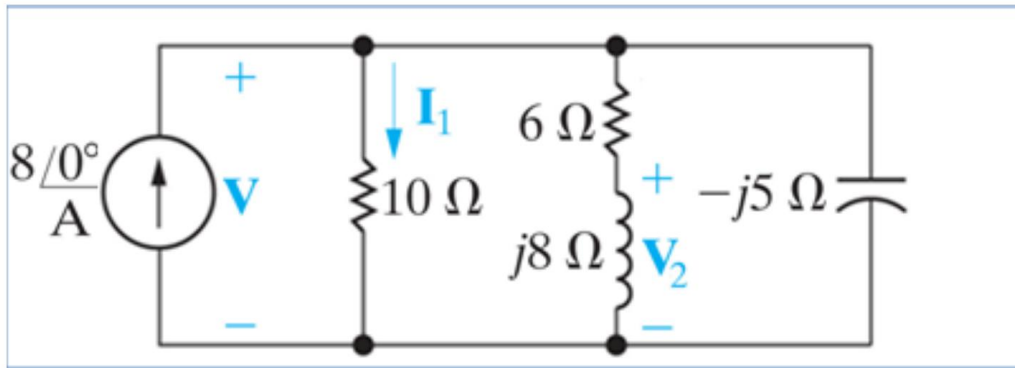
Example 9.9: Step 6, $v(t)$

Example 9.9

$i_s(t) = 8 \cos(200,000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady state.



6. Inverse phasor-transform the result to get back to the time domain.

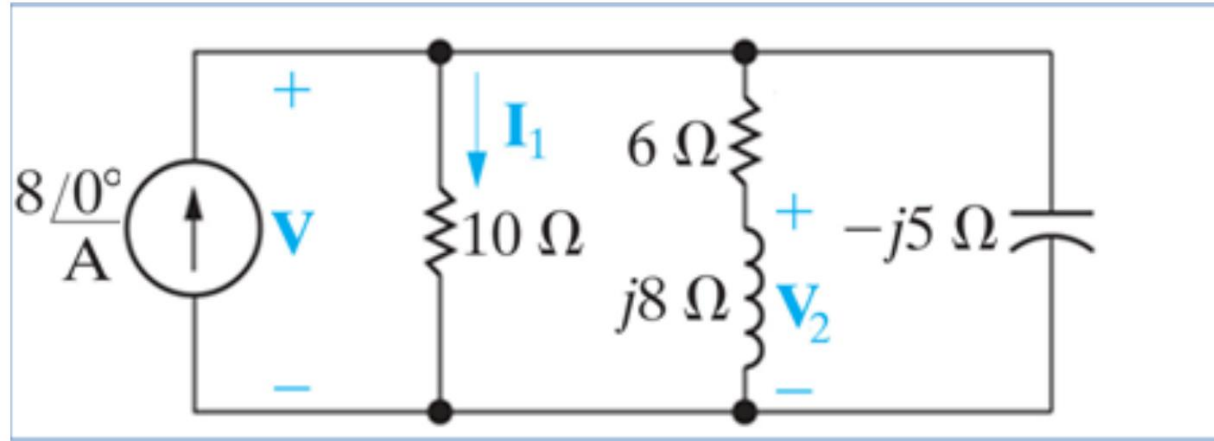


$$\mathbf{V} = (32 - j24)\text{V} = 40 \angle -36.87^\circ \text{V}$$

$$v(t) = \mathcal{P}^{-1}\{40 \angle -36.87^\circ\} = 40 \cos(200,000t - 36.87^\circ) \text{V}$$

Example 9.9: Check Your Understanding (3)

The most direct way to calculate the phasor I_1 is

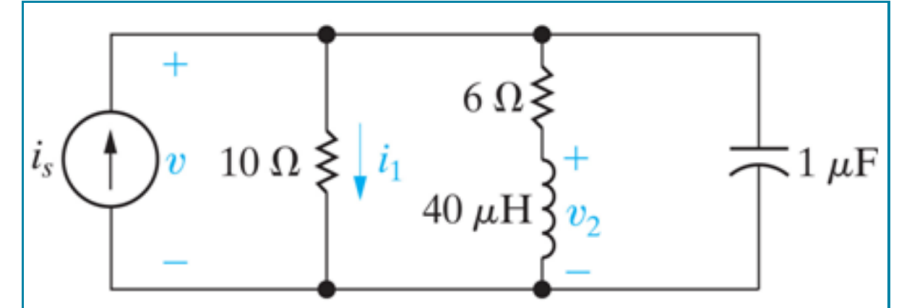


- ☒ A. Voltage division
- ☒ B. Current division
- ☒ C. Source transform

Example 9.9: Step 5, Find I_1

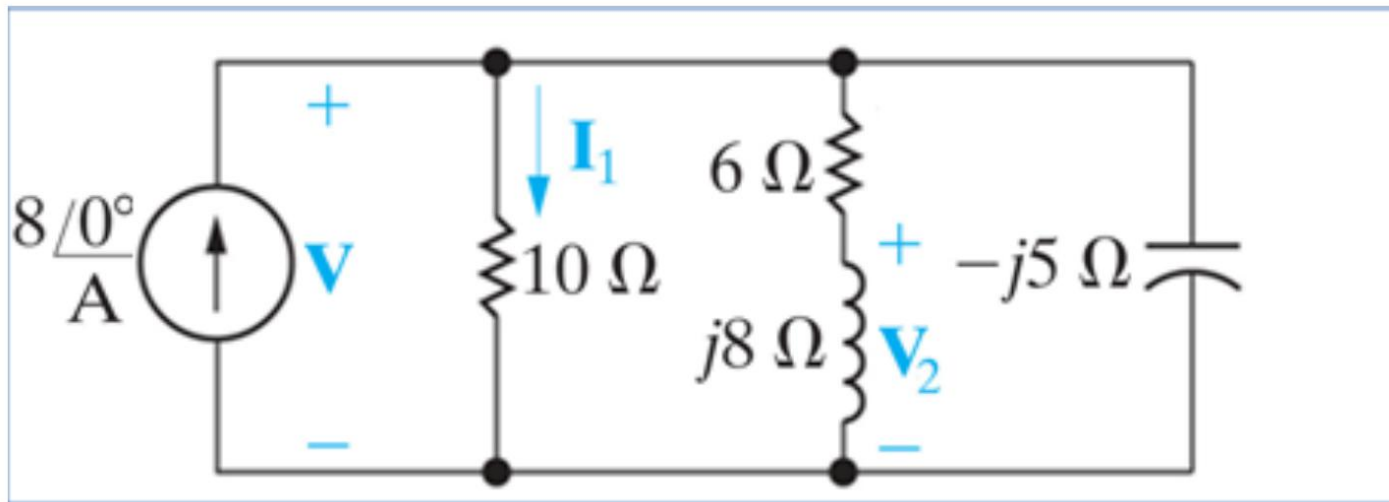
Example 9.9

$i_s(t) = 8 \cos(200,000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady state.



5. Use DC circuit analysis techniques to find the phasor I_1 .

– Use current division.

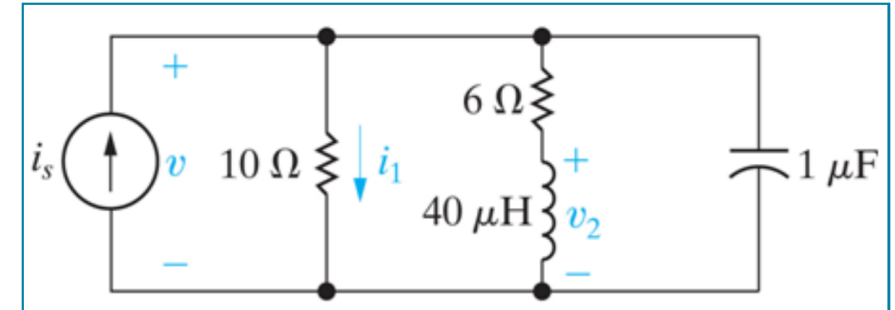


$$I_1 = \frac{Z_{eq}}{Z_1} I_s = \frac{(4 - j3)}{10} (8 \angle 0^\circ) = (3.2 - j2.4) \text{ A}$$

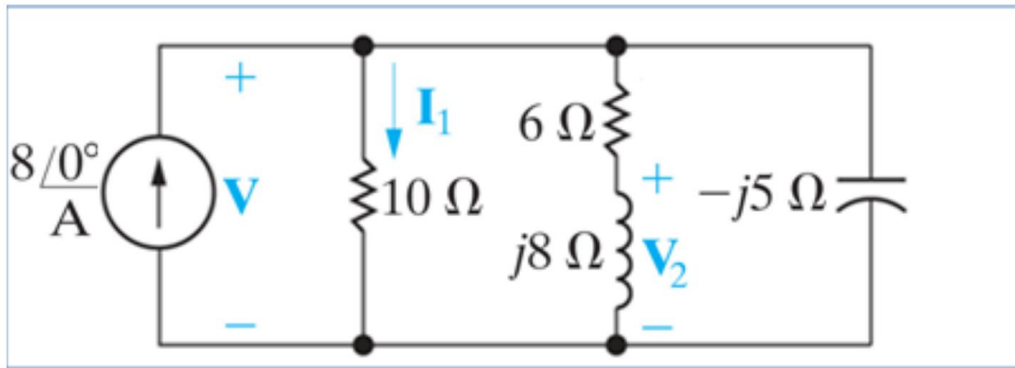
Example 9.9: Step 6, $i_1(t)$

Example 9.9

$i_s(t) = 8 \cos(200,000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady state.



6. Inverse phasor-transform the result to get back to the time domain.

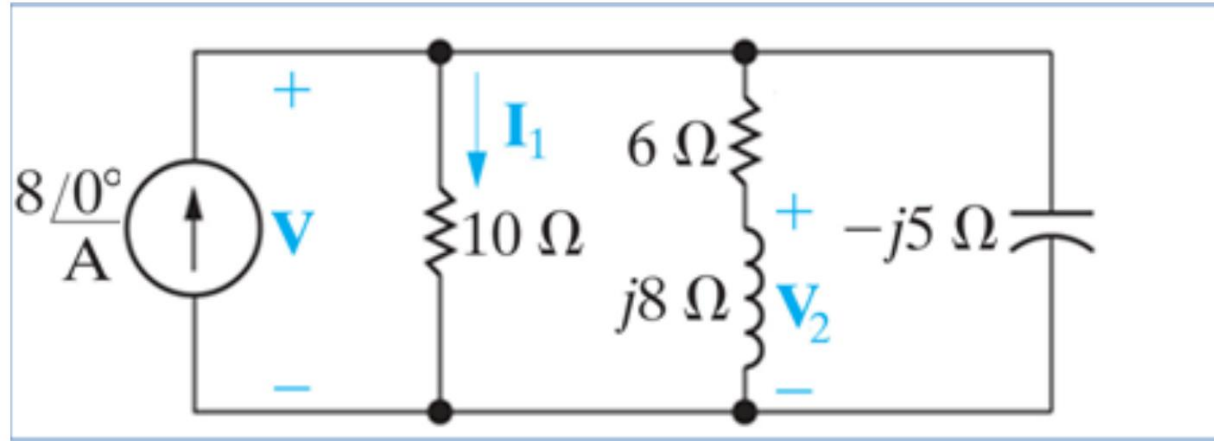


$$I_1 = (3.2 - j2.4) \text{ A} = 4 \angle -36.87^\circ \text{ A}$$

$$i_1(t) = \mathcal{P}^{-1}\{4 \angle -36.87^\circ\} = 4 \cos(200,000t - 36.87^\circ) \text{ A}$$

Example 9.9: Check Your Understanding (4)

The most direct way to calculate the phasor V_2 is

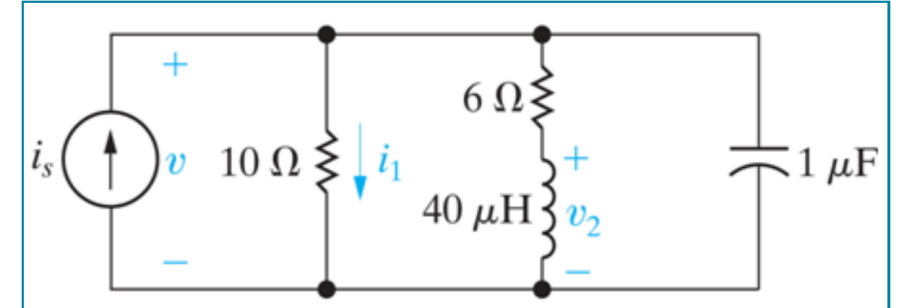


- ✓ A. Voltage division
- ✗ B. Current division
- ✗ C. Node voltage method

Example 9.9: Step 5, Find V_2

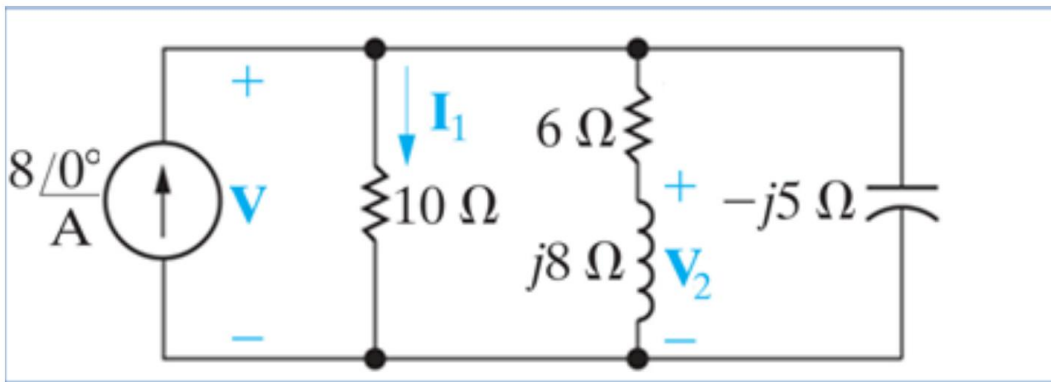
Example 9.9

$i_s(t) = 8 \cos(200,000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady state.



5. Use DC circuit analysis techniques to find the phasor V_2 .

– Use voltage division.

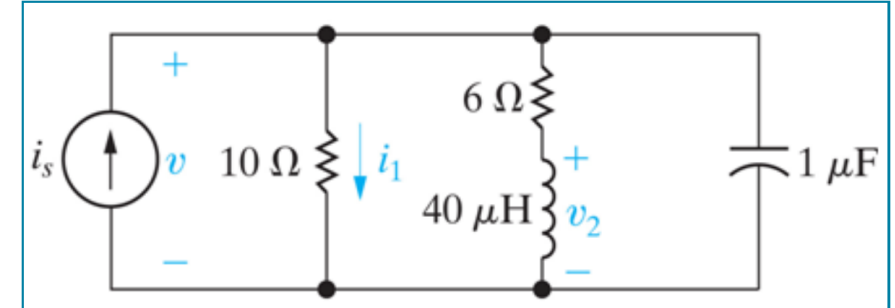


$$\begin{aligned} \mathbf{V}_2 &= \frac{Z_L}{Z_R + Z_L} \mathbf{V} = \frac{j8}{6 + j8} (32 - j24) \\ &= (32 + j0) \text{ V} \end{aligned}$$

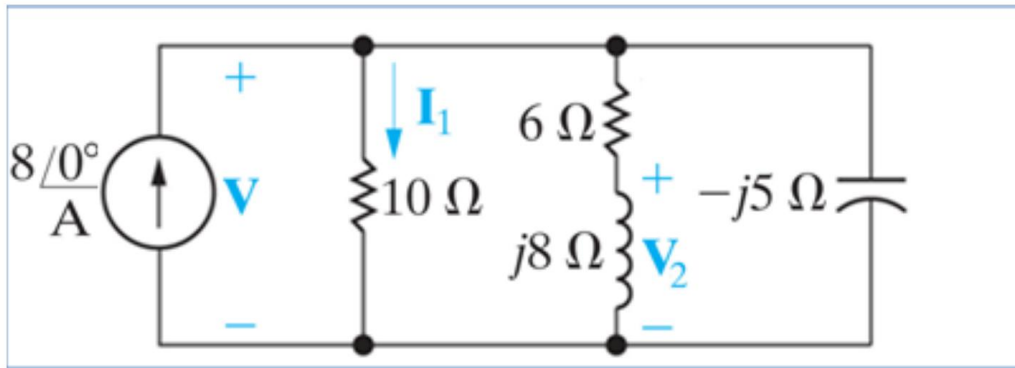
Example 9.9: Step 6, $v_2(t)$

Example 9.9

$i_s(t) = 8 \cos(200,000t)$ A. Find $v(t)$, $i_1(t)$, and $v_2(t)$ in the steady state.



6. Inverse phasor-transform the result to get back to the time domain.

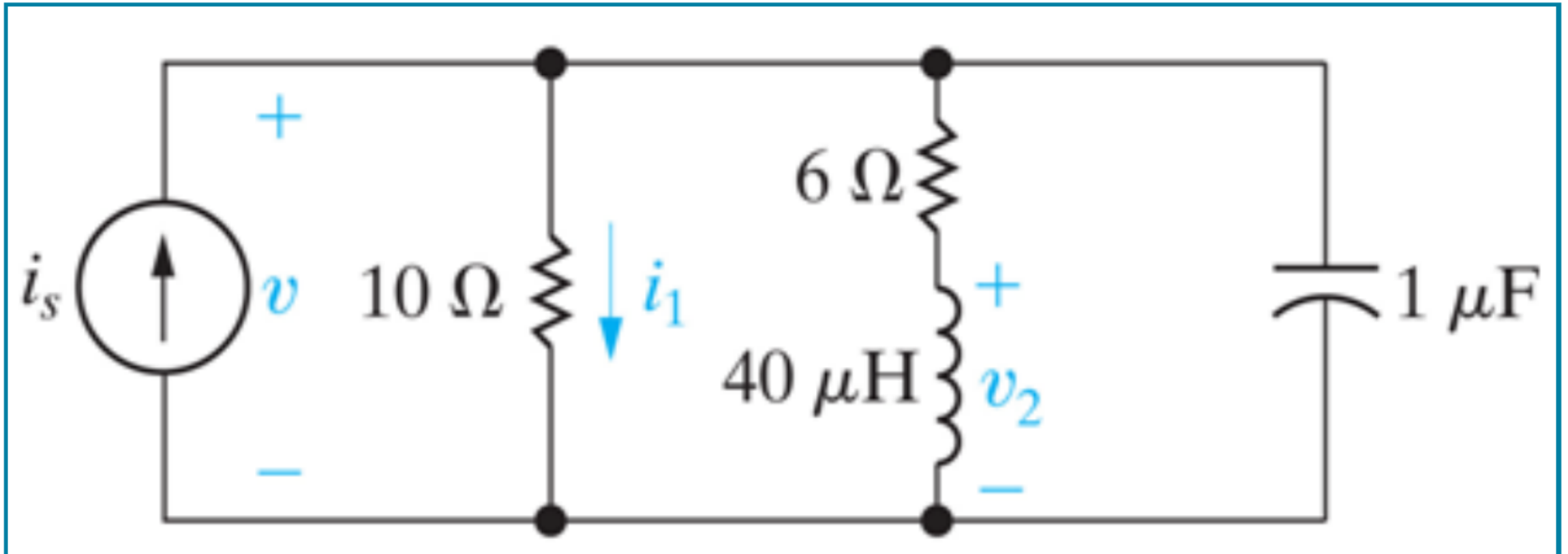


$$\mathbf{V}_2 = (32 + j0) \text{ V} = 32 \angle 0^\circ \text{ V}$$

$$v_2(t) = \mathcal{P}^{-1}\{32 \angle 0^\circ\} = 32 \cos 200,000t \text{ V}$$

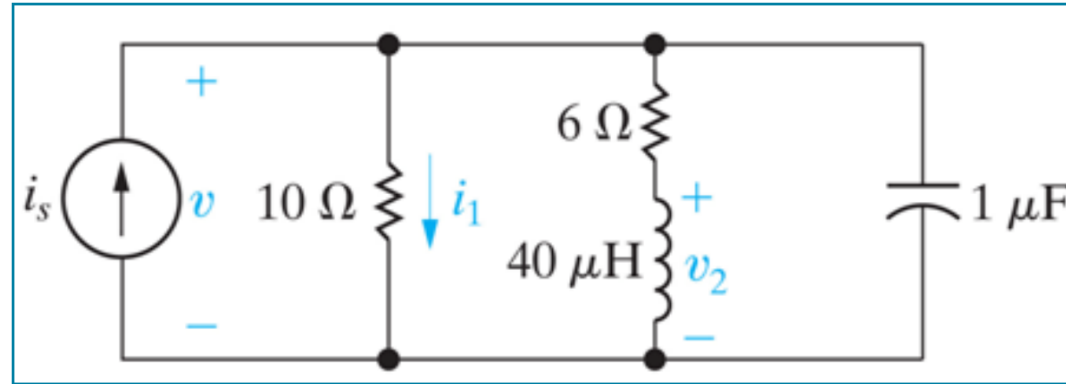
Example 9.9: A Frequency Question

Suppose we can vary the frequency of the current source. What frequency will cause $i_s(t)$ and $v(t)$ to be in phase in the steady state?



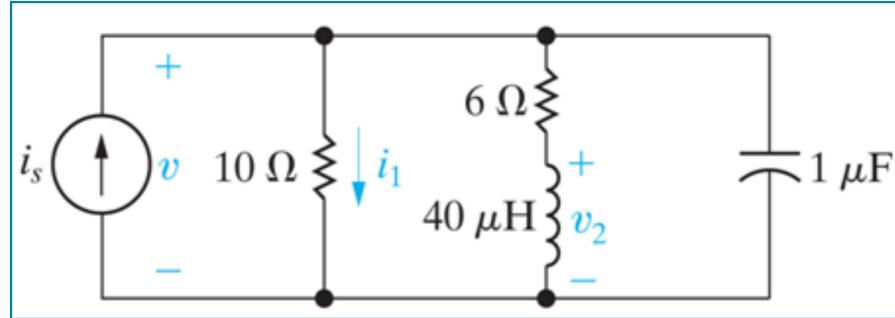
Example 9.9: Check Your Understanding (5)

If $i_s(t)$ and $v(t)$ are in phase in the steady state, this means



- ☒ A. Their phase angles are the same.
- ☒ B. The equivalent impedance seen by the current source has a phase angle of 0° .
- ☒ C. The equivalent impedance seen by the current source is purely resistive.
- ☒ D. All of the above.

Example 9.9: Solving for the Resonant Frequency

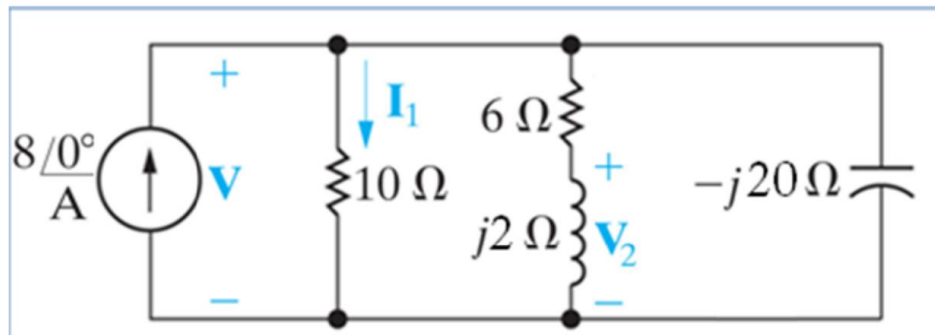
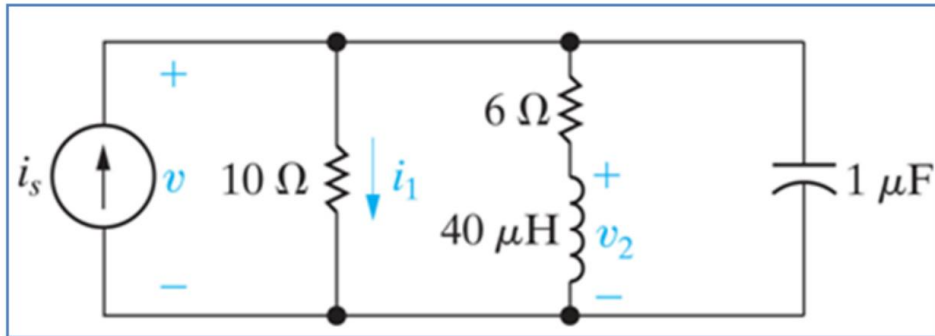


Set the imaginary part of the equivalent admittance to zero and solve for ω .

$$\begin{aligned}\frac{1}{Z_{eq}} &= \frac{1}{10} + \frac{1}{6 + j\omega L} + j\omega C \\ &= \frac{(6 + j\omega L)(6 - j\omega L)}{10(6 + j\omega L)(6 - j\omega L)} + \frac{10(6 - j\omega L)}{10(6 + j\omega L)(6 - j\omega L)} + \frac{10(j\omega C)(6 + j\omega L)(6 - j\omega L)}{10(6 + j\omega L)(6 - j\omega L)} \\ &= \frac{36 + \omega^2 L^2 + 60 - 10j\omega L + 10j\omega C(36 + \omega^2 L^2)}{10(36 + \omega^2 L^2)} \\ \mathcal{Im}\left\{\frac{1}{Z_{eq}}\right\} &= \frac{-10\omega L + 10\omega C(36 + \omega^2 L^2)}{10(36 + \omega^2 L^2)} = 0 \quad \therefore \quad -10\omega L + 10\omega C(36 + \omega^2 L^2) = 0 \\ \therefore \quad \omega^2 &= \frac{L/C - 36}{L^2} = \frac{40/1 - 36}{(40\mu)^2} \quad \therefore \quad \omega = 50,000 \text{ rad/s}\end{aligned}$$

Example 9.9: Verification at $\omega = 50,000$ rad/s

Suppose we can vary the frequency of the current source. What frequency will cause $i_s(t)$ and $v(t)$ to be in phase in the steady state?



For $\omega = 50,000$ rad/s

$$Z_L = j\omega L = j(50,000)(40\mu) = j2\Omega$$

$$Z_C = \frac{-j}{\omega C} = \frac{-j}{(50,000)(1\mu)} = -j20\Omega$$

$$\therefore Z_{eq} = 10 \parallel (6 + j2) \parallel -j20 = 4\Omega$$

Homework

Rizzoni, Principles and Applications of Electrical Engineering, 7th ed., Chapter 3

- 3.2, 3.7, 3.14, 3.17
 - capacitor and inductor i-v relations, energy
- 3.43, 3.47, 3.52
 - phasors, equivalent impedance, phasor solution of a circuit